

Agentic AI-Based Dynamic Tariff Optimization for EV Charging Networks

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Executive Summary

- **Data Foundation:** Analyzed 2 real-world EV datasets – ACN-Data (14,999 sessions, Caltech/JPL) and UrbanEV ST-EVCDP (24,798 piles, Shenzhen)
- **3-Agent Pipeline:** Demand Prediction Agent (XGBoost + Random Forest), Tariff Pricing Agent (surge Rs.19.5/kWh, discount Rs.12/kWh), and Monitoring & Learning Agent with autonomous threshold adjustment
- **Key Results:** 7.34% utilization improvement + 15% off-peak volume uplift – demand successfully redistributed from peak to off-peak windows
- **Impact:** Replicable, data-driven alternative to flat Rs.15/kWh pricing for operators, grid managers, and policymakers

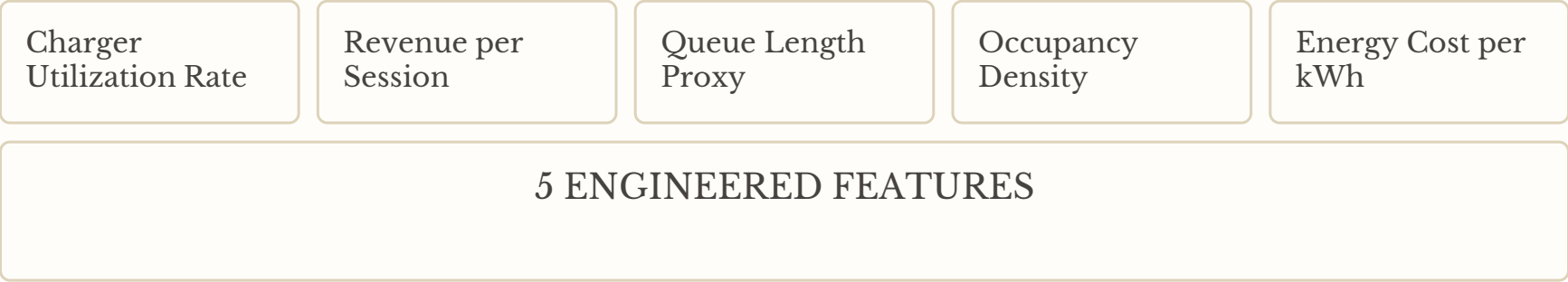
Data Foundation

Two Real-World Datasets

Dataset	ACN-Data	UrbanEV
Sessions / Piles	14,999	24,798 piles
Granularity	Per-session	5-min intervals
Location	Caltech/JPL, USA	Shenzhen, China
Use	User behavior, overstay	Demand, pricing

Preprocessing & Engineered Features

- Dropped 1,305 ACN sessions with missing fields (connectionTime, disconnectTime, kWh)
- Filled 8 doneChargingTime gaps → filled with disconnectTime
- Corrected 18 negative durations → replaced with session duration
- UrbanEV: zero missing values.



Key EDA Findings & Demand Behavior Insights

Temporal Patterns

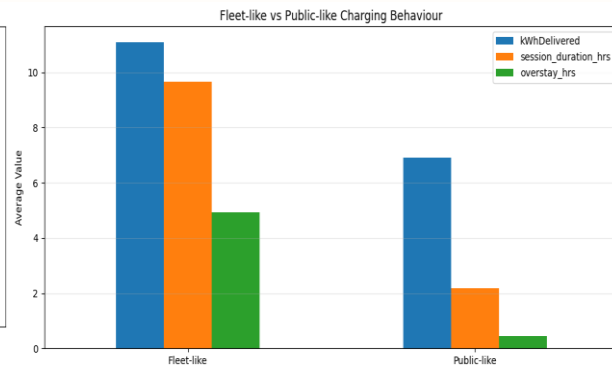
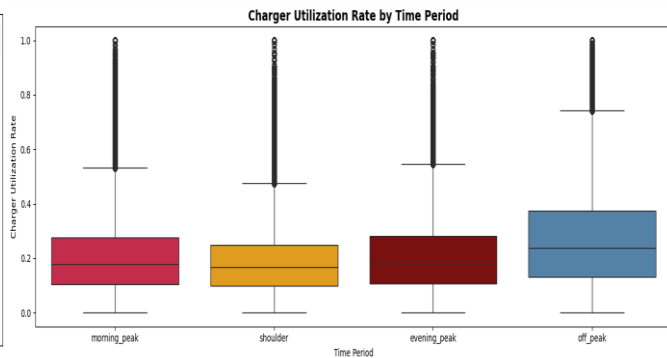
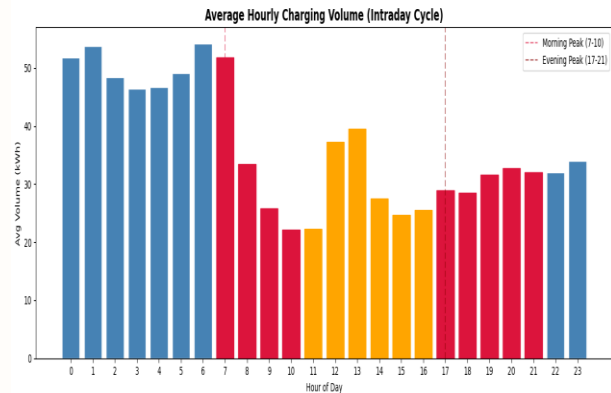
- Two peak windows: **Morning 7–10am** and **Evening 5–9pm**.
- Weekday demand consistently higher; evening peaks sharper on weekdays.
- Stable weekly seasonality, no long-run decline.

Utilization & Congestion

- CBD stations more utilized across all hours.
- Peak utilization variance highest during morning and evening windows box plot shows wider spread vs shoulder/off-peak
- Dynamic pricing stations already show higher revenue per slot vs. static pricing stations

User Behavior (ACN)

- Avg session ~3–4 hrs; significant **overstay** (1+ hr after charging completes).
- Fleet-like vs. public-like users show distinct usage signatures.
- Overstay is a key congestion driver – justifies time-based surge pricing.



Demand Prediction Agent

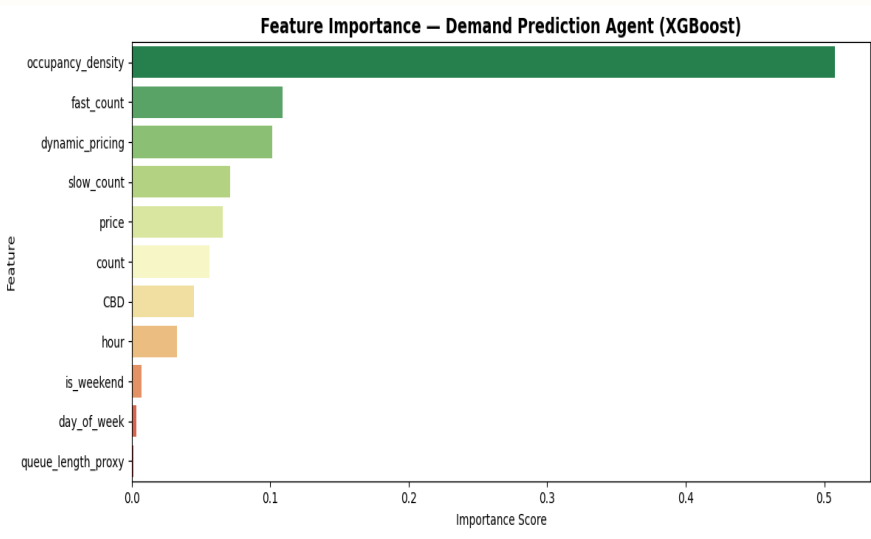
Architecture

- **3 outputs** from **11 features** from UrbanEV: hour, day_of_week, is_weekend, count, fast_count, slow_count, CBD, dynamic_pricing, occupancy_density, queue_length_proxy, price.
- 80/20 train-test split, random_state=42.

Model Results

Output	Model	Key Metric
Charger Utilization Rate	XGBoost Regressor	RMSE: 0.0285 MAE: 0.0183 R²: 0.9681
Expected Charging Load	XGBoost Regressor	RMSE: 13.4637 MAE: 4.7843 R²: 0.9825
Congestion Probability	Random Forest Classifier	Avg Congestion Probability: 0.0063 F1: 0.97 Precision: 0.96 Recall: 0.97

Top Feature Importance

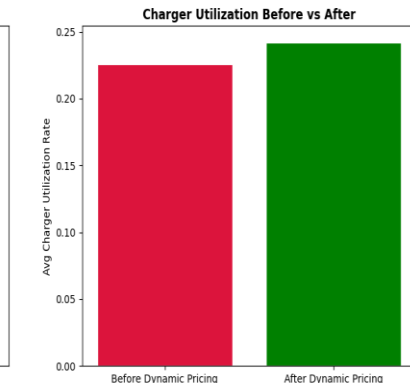
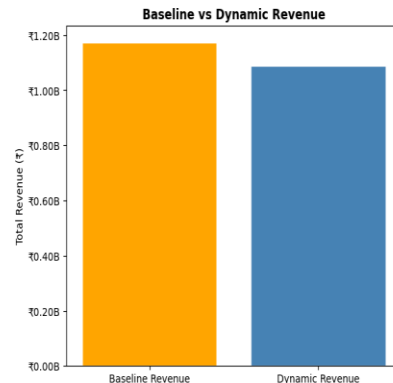
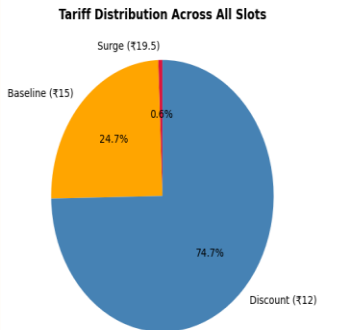


The model reliably identifies high-utilization windows, enabling the Tariff Agent to act **before** congestion occurs.

Tariff Pricing Agent

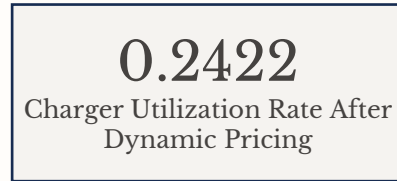
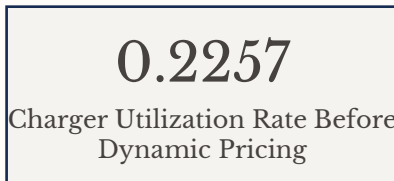
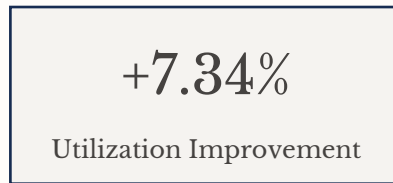
3-Tier Pricing Logic

Condition	Tariff	Rationale
Predicted Utilization > 80%	Rs. 19.5/kWh (+30%)	Surge – deter marginal users
Predicted Utilization < 30%	Rs. 12/kWh (~20%)	Discount – attract off-peak demand
Otherwise	Rs. 15/kWh (baseline)	Neutral – no action



Tariff Distribution

- Surge slots: 12,854 (0.61%)
- Baseline slots: 527,249 (25.07%)
- Discount slots: 1,593,977 (74.32%)
- Improvement in Charger Utilization Rate : 7.34%
- Revenue decrease reflects the cost of incentivizing off-peak charging through discounted tariffs.

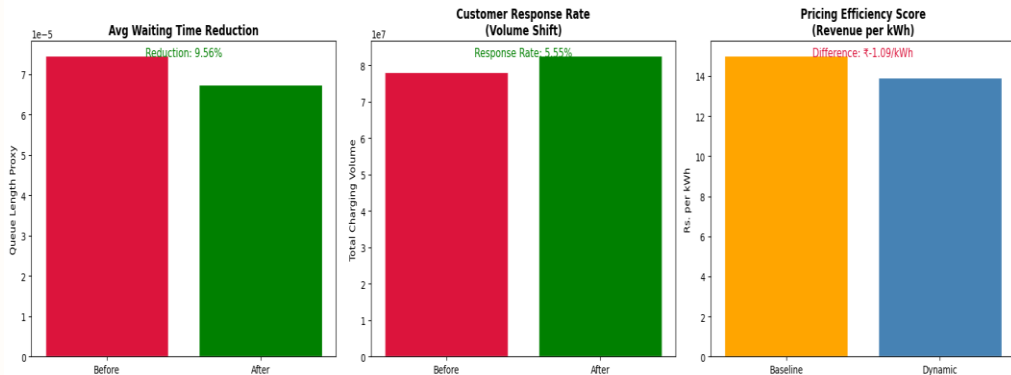


Monitoring & Learning Agent

What It Evaluates

Metric	Before	After
Avg Waiting Time (queue proxy)	0.000075	0.000067
Customer Response Rate	—	5.55%
Pricing Efficiency Score (₹/kWh)	Rs. 15/kWh	Rs. 13.9125/kWh

- Daily revenue tracking (Dynamic vs. Baseline) confirms the trade-off is consistent, not erratic.
- Waiting Time Reduction = 9.56%



Autonomous Feedback Loop

Pricing Efficiency < Baseline

→ Flag for threshold review

Waiting Time Reduction ≤ 0

→ Tighten surge threshold: 80% → 75%

Customer Response Rate ≤ 0

→ Widen discount threshold: 30% → 35%

Key finding:

- Congestion reduced at peak hours and customers respond to tariff changes – foundation for reinforcement learning in future iterations.
- Pricing efficiency dip to ₹13.91/kWh is expected – 74.32% of slots are discount slots, deliberately trading short-run revenue for long-run utilization and demand redistribution.

Business, Operational & Policy Implications

For Operators

- Replace flat ₹15/kWh with 3-tier dynamic pricing – implementable with existing occupancy data
- Focus surge on CBD stations during 7–10am & 5–9pm – highest ROI
- Off-peak discounts drive +15% volume uplift without adding infrastructure

For Grid Management

- Dynamic pricing smooths demand curves – reduces peak load spikes
- +7.34% utilization achieved without adding chargers

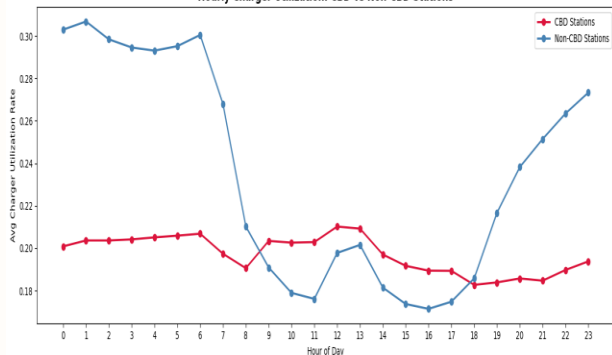
For Policy

- EDA suggests dynamic pricing stations outperform static stations in revenue per slot – evidence for time-of-use EV tariff regulation
- Overstay is a major congestion driver – consider overstay penalty pricing
- Framework generalizable: built on Shenzhen + US data, applicable to Indian EV networks

 **Limitations:** Revenue simulation uses assumed elasticity (–10% surge, +15% discount uptake); ACN data not in ML pipeline; feedback loop is rule-based, not reinforcement learning.

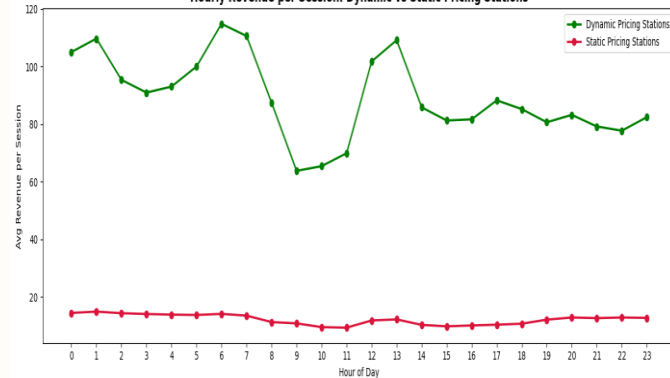
Appendix A1 - Station-Level Demand & Revenue Patterns

Hourly Charger Utilization: CBD vs Non-CBD Stations

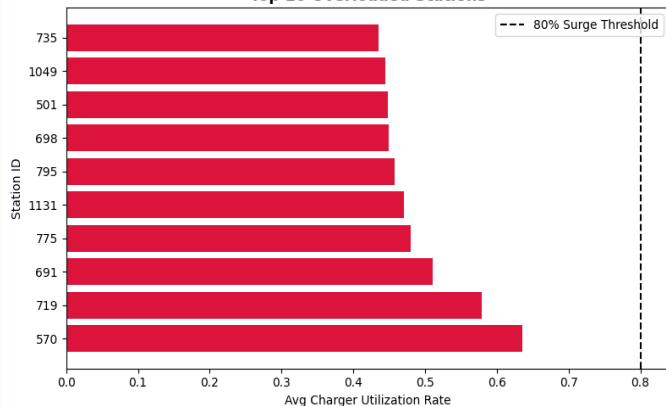


- CBD stations consistently more utilized-gap widens at peak hours.
- Dynamic pricing stations already outperform static in revenue per slot
- Top overloaded stations: all CBD, peak-hour concentrated.
- Top underutilized: all non-CBD, off-peak

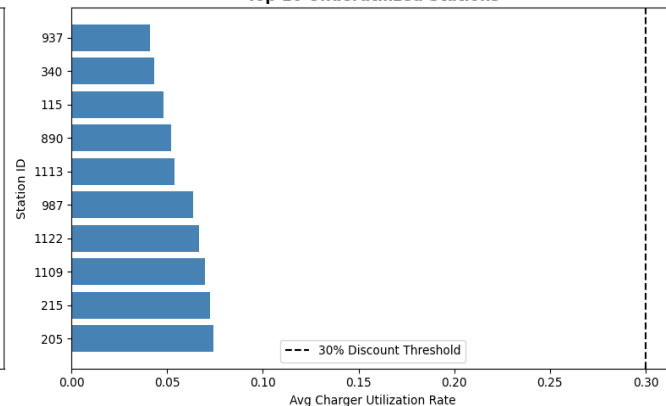
Hourly Revenue per Session: Dynamic vs Static Pricing Stations



Top 10 Overloaded Stations

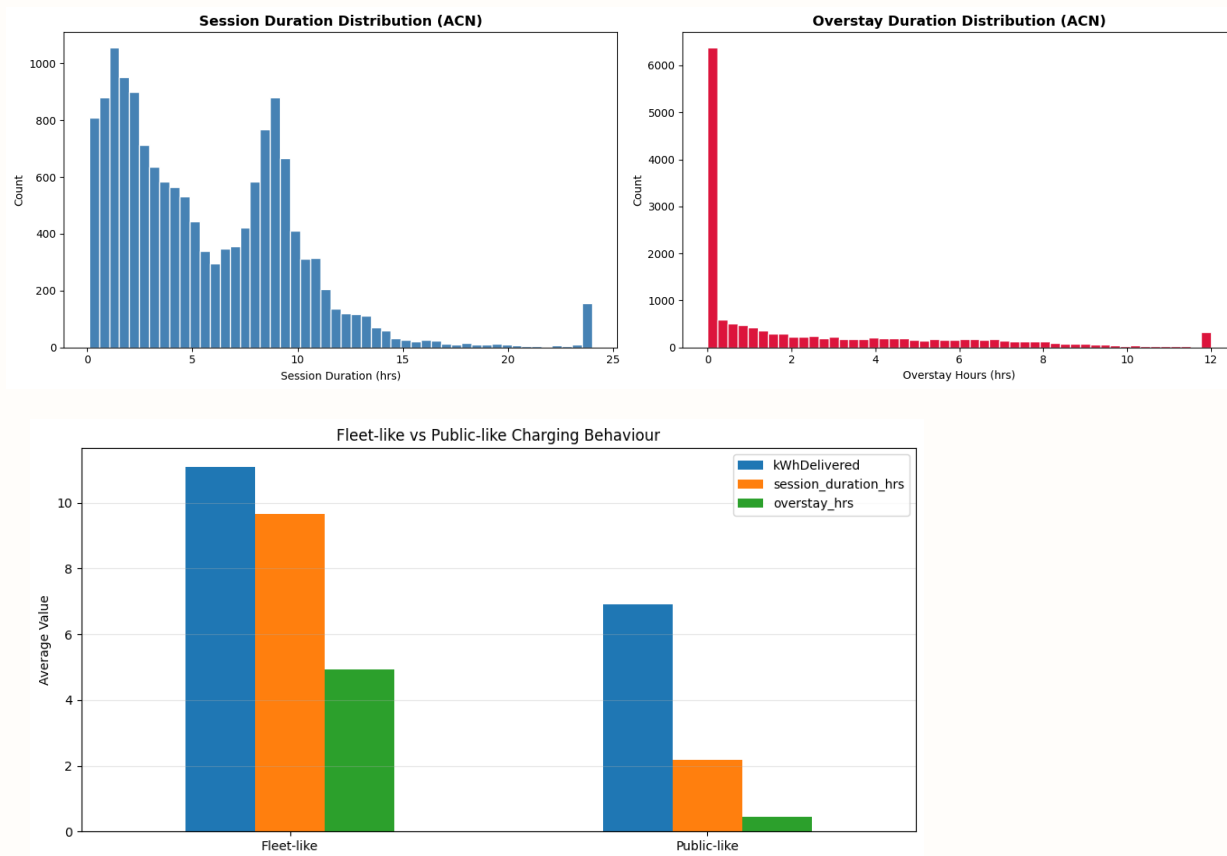


Top 10 Underutilized Stations



Appendix A2 - ACN Behavior: Session Behavior & Overstay Analysis

- Avg session duration ~3–4 hrs; overstay begins well before disconnect.
- Sessions with overstay >1 hr silently block chargers during peak periods.
- Fleet-like vs Public-like users have distinct energy and duration signatures.



Appendix A3 - Assumptions & Limitations

- Elasticity values (-10% surge deterrence, +15% discount uptake) are assumed, not empirically estimated, impacting revenue projections.
- The feedback loop for the Monitoring & Learning Agent is currently rule-based, meaning thresholds adjust once and not iteratively through reinforcement learning.
- ACN dataset was used only for Exploratory Data Analysis (EDA) and was not integrated into the Machine Learning (ML) pipeline for demand prediction.
- UrbanEV data is sourced from Shenzhen, requiring local calibration and validation for optimal applicability to diverse Indian EV charging networks.